

Smoothing Techniques in Image Processing

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Recapitulation

❖ Point Processing Techniques

- Image Negatives
- Logarithmic transformation for Dynamic range Compression.
- Power Law transformation
- Contrast Stretching
- Gray Level Slicing

❖ Histogram Based Processing Techniques

- Histogram Equalization
- ❖ Histogram Modification/Specification Image Differencing
- ❖ Image Averaging

Content

❑ On completion Student will be able to understand

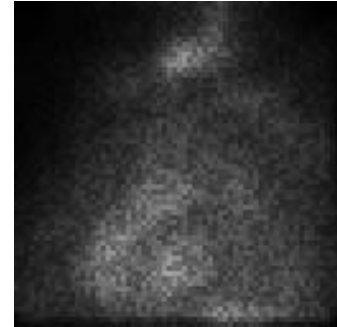
- ❖ Introduction
- ❖ Mask Processing Techniques
- ❖ Brief review of linear operators
- ❖ Linear image smoothing techniques
- ❖ Nonlinear/Median image smoothing techniques

Introduction

Why do we need image smoothing?

What is “image” and what is “noise”?

- Frequency spectrum
- Statistical properties



Local or Neighborhood Operation

- ❖ In neighborhood operation, the pixels in an image are modified based on some function of the pixels in their neighborhood.
- ❖ Linear spatial filtering is often referred to as convolving a mask with an image.
- ❖ The filter masks are sometimes called convolution masks or convolution kernels.
- ❖ Spatial Filtering
 - Spatial filtering involves passing a weighted mask or kernel over the image and replacing the original image pixel value corresponding to the Center of the kernel with the sum of the original pixel values in the region corresponding to the kernel multiplied by the kernel weights.

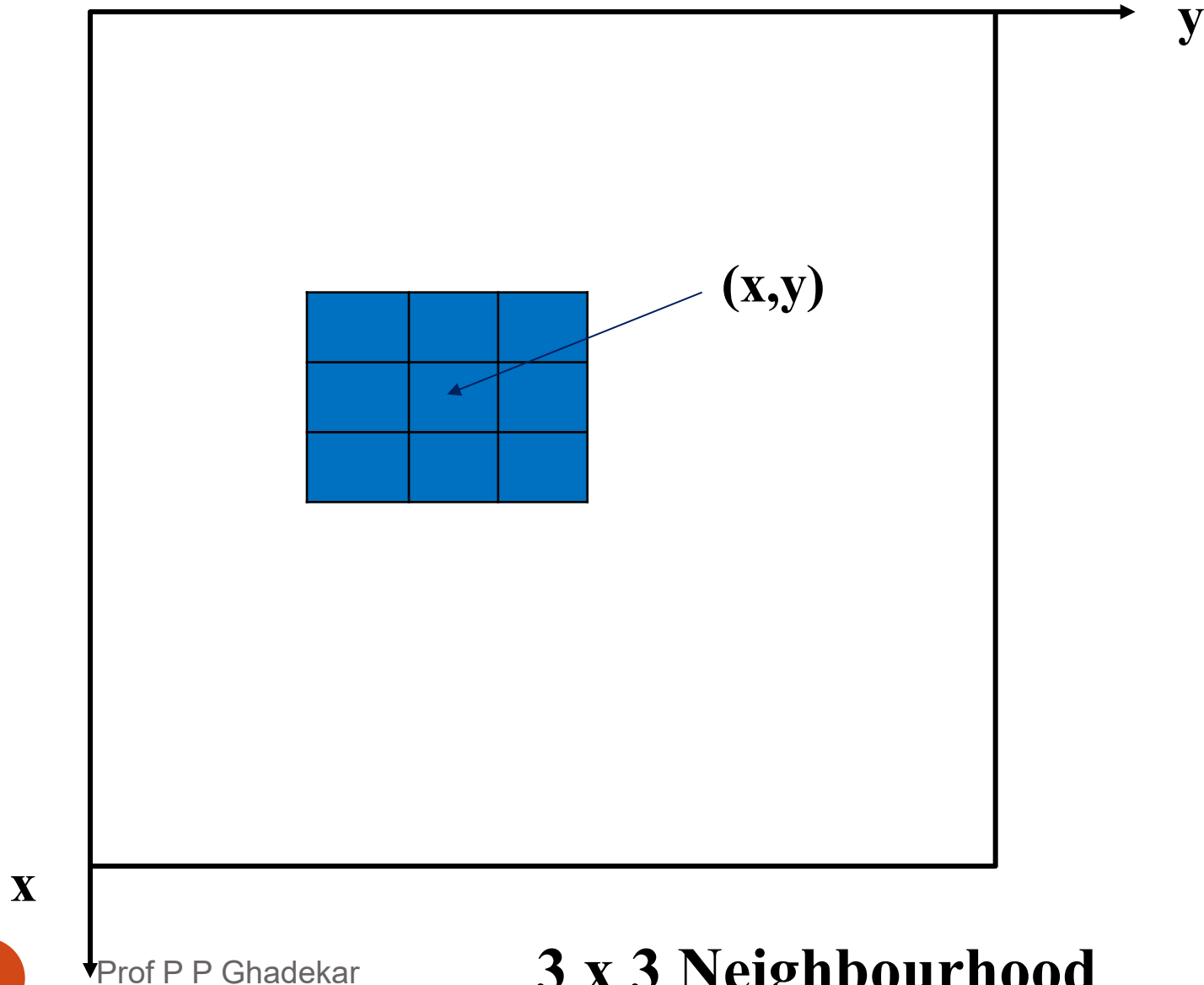
Linear Filtering

- ❖ In linear filtering , each pixel in the input image is replaced by a linear combination of intensities of neighboring pixels.
- ❖ That is each pixel value in the output image is a weighted sum of the pixels in the neighborhood of the corresponding pixel in the input image.
- ❖ Linear filtering can be used to smoothen an image as well as sharpen the image.
- ❖ A spatially invariant linear filter can be implemented using a convolution mask.
- ❖ If different filter weights are used for different parts of the image then the linear filter is a spatially varying.

Mean Filter or Average Filter or Low pass Filter

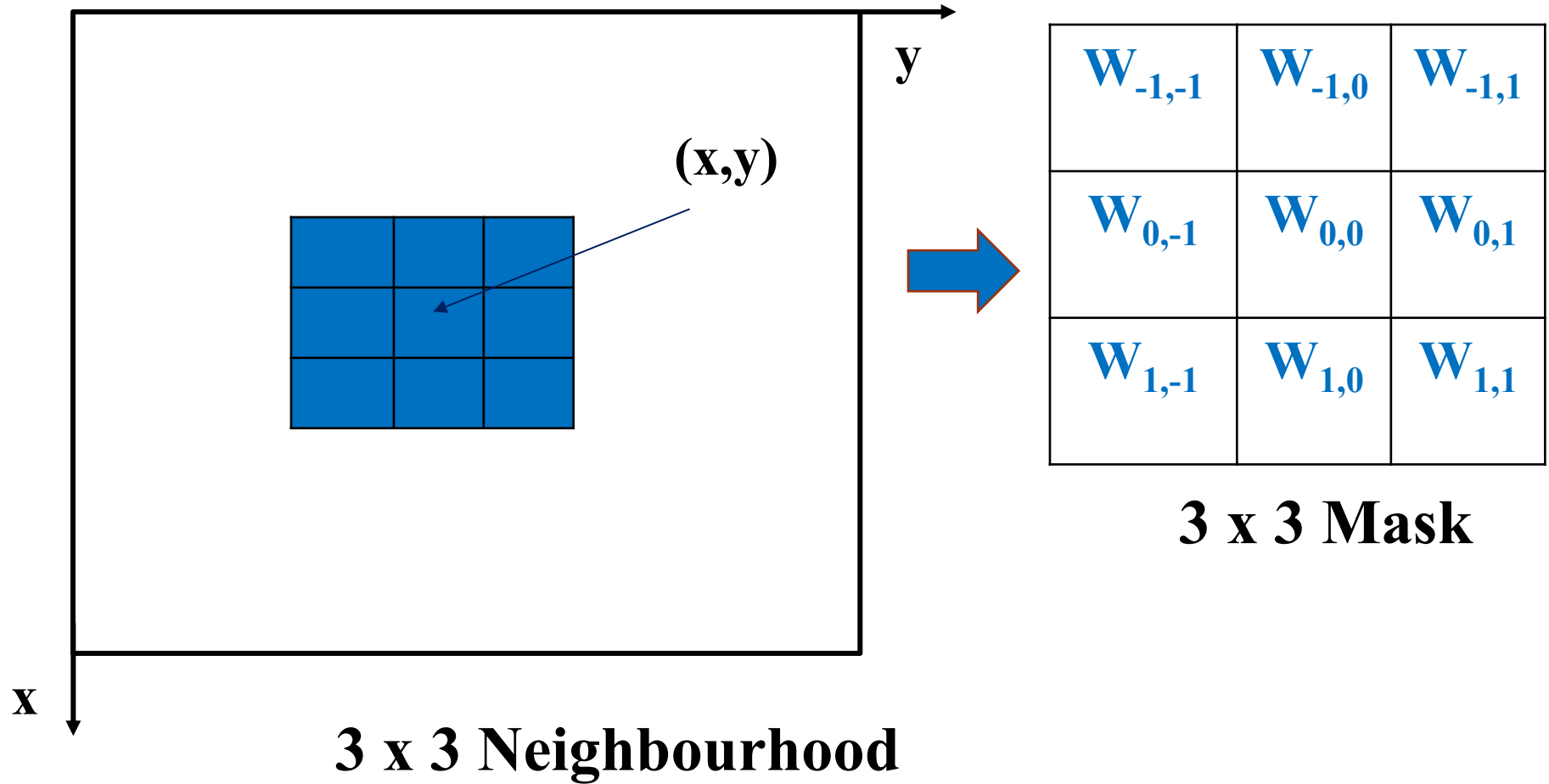
- ❖ The mean filter replaces each pixel by the average of all the values in the neighborhood. The size of the neighborhood controls the amount of filtering.
- ❖ In a spatial averaging operation, each pixel is replaced by a weighted average of its neighborhood pixels.
- ❖ The low pass filter preserves the smooth region in the image and removes the sharp variations leading to blurring effect.

Neighbourhood



3 x 3 Neighbourhood

Mask Processing



$$g(x,y) = \sum_{i=-1}^1 \sum_{j=-1}^1 W_{i,j} f(x+i, y+j)$$

Smoothing Spatial Filter

$$\frac{1}{9} \times \begin{array}{|c|c|c|} \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline 1 & 1 & 1 \\ \hline \end{array}$$

Smoothing Spatial Filter

Averaging Filter (low pass Filter)

1/9

1	1	1
1	1	1
1	1	1

**3 x 3 Averaging
Mask**



Box Filter

$$g(x,y) = \frac{1}{9} \sum_{i=-1}^1 \sum_{j=-1}^1 f(x+i, y+j)$$

Illustration of Spatial Filtering-Example

7	9	11
10	50	8
9	5	6

Original Image

1/9

1	1	1
1	1	1
1	1	1

**3 x 3 Averaging
Mask**

0	0	0	0	0
0	7	9	11	0
0	10	50	8	0
0	9	5	6	0
0	0	0	0	0

Input Image after zero padding

Illustration of Spatial Filtering-Example

1/9 0	1/9 0	1/9 0	0	0
1/9 0	1/9 7	1/9 9	11	0
1/9 0	1/9 10	1/9 50	8	0
0	9	5	6	0
0	0	0	0	0

$$0 \times 1/9 + 0 \times 1/9 + 0 \times 1/9 + 0 \times 1/9 + 7 \times 1/9 + 9 \times 1/9 + 0 \times 1/9 + 10 \times 1/9 + 50 \times 1/9 = 8.4$$

Illustration of Spatial Filtering-Example

0	$\frac{1}{9}$ 0	$\frac{1}{9}$ 0	$\frac{1}{9}$ 0	0
0	$\frac{1}{9}$ 8.4	$\frac{1}{9}$ 9	$\frac{1}{9}$ 11	0
0	$\frac{1}{9}$ 10	$\frac{1}{9}$ 50	$\frac{1}{9}$ 8	0
0	9	5	6	0
0	0	0	0	0

$$0 \cdot \frac{1}{9} + 0 \cdot \frac{1}{9} + 0 \cdot \frac{1}{9} + 7 \cdot \frac{1}{9} + 9 \cdot \frac{1}{9} + 11 \cdot \frac{1}{9} + 10 \cdot \frac{1}{9} + 50 \cdot \frac{1}{9} + 8 \cdot \frac{1}{9} = 10.5$$

Illust

ample

0	0	$\frac{1}{9}$ 0	$\frac{1}{9}$ 0	$\frac{1}{9}$ 0
0	8.4	$\frac{1}{9}$ 10.5	$\frac{1}{9}$ 11	$\frac{1}{9}$ 0
0	10	$\frac{1}{9}$ 50	$\frac{1}{9}$ 8	$\frac{1}{9}$ 0
0	9	5	6	0
0	0	0	0	0

$$0 \cdot \frac{1}{9} + 0 \cdot \frac{1}{9} + 0 \cdot \frac{1}{9} + 9 \cdot \frac{1}{9} + 11 \cdot \frac{1}{9} + 0 \cdot \frac{1}{9} + 50 \cdot \frac{1}{9} + 8 \cdot \frac{1}{9} + 0 \cdot \frac{1}{9} = 8.66$$

Illustration of Spatial Filtering-Example

0	0	0	0	0
$\frac{1}{9}$ 0	$\frac{1}{9}$ 8.4	$\frac{1}{9}$ 10.7	8.6	0
$\frac{1}{9}$ 0	$\frac{1}{9}$ 10	$\frac{1}{9}$ 50	8	0
$\frac{1}{9}$ 0	$\frac{1}{9}$ 9	$\frac{1}{9}$ 5	6	0
0	0	0	0	0

$$0 \cdot \frac{1}{9} + 7 \cdot \frac{1}{9} + 9 \cdot \frac{1}{9} + 0 \cdot \frac{1}{9} + 10 \cdot \frac{1}{9} + 50 \cdot \frac{1}{9} + 0 \cdot \frac{1}{9} + 9 \cdot \frac{1}{9} + 5 \cdot \frac{1}{9} = 10.0$$

Illustration of Spatial Filtering-Example

0	0	0	0	0
0	8.4444	10.5556	8.6667	0
0	10.0000	12.7778	9.8889	0
0	8.2222	9.7778	7.6667	0
0	0	0	0	0

Result of Averaging Filter

7	9	11
10	50	8
9	5	6

Original Image

8.4444	10.5556	8.6667
10.0000	12.7778	9.8889
8.2222	9.7778	7.6667

Image after Spatial Averaging

Spatial Low Pass Filter-Example



Original Image

3×3
Smoothing
filter



Smoothened Image

Spatial Low Pass Filter-Example



Original Image

5×5
Smoothing
filter



Smoothened Image

Spatial Low Pass Filter-Example



FIGURE (a) Original image, of size 500*500 pixels. (b)–(f) Results of smoothing with square averaging filter masks of sizes $n=3, 5, 9, 15,$ and 35 , respectively

Linear Image smoothing techniques

Box filters. Arithmetic mean $L \times L$ operator

❖ An example

$$\begin{array}{cccccc}
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 \dots & 8 & 5 & 8 & 8 & 5 & 8 & \dots \\
 \dots & 8 & 5 & 8 & 8 & 5 & 8 & \dots \\
 \dots & 8 & 5 & 8 & 8 & 5 & 8 & \dots \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & &
 \end{array}
 * \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{array}{cccccc}
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 \dots & 7 & 7 & 7 & 7 & 7 & 7 & \dots \\
 \dots & 7 & 7 & 7 & 7 & 7 & 7 & \dots \\
 \dots & 7 & 7 & 7 & 7 & 7 & 7 & \dots \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & &
 \end{array}$$

$$\begin{array}{cccccc}
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 \dots & 8 & 5 & 8 & 5 & 8 & 5 & \dots \\
 \dots & 8 & 5 & 8 & 5 & 8 & 5 & \dots \\
 \dots & 8 & 5 & 8 & 5 & 8 & 5 & \dots \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & &
 \end{array}
 * \frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{array}{cccccc}
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\
 \dots & 6 & 7 & 6 & 7 & 6 & 7 & \dots \\
 \dots & 6 & 7 & 6 & 7 & 6 & 7 & \dots \\
 \dots & 6 & 7 & 6 & 7 & 6 & 7 & \dots \\
 \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & &
 \end{array}$$

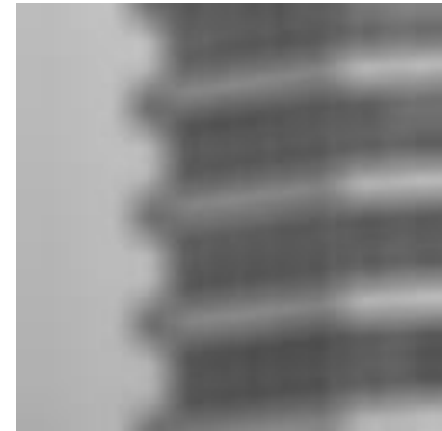
Linear Image smoothing techniques

Box filters. Arithmetic mean $L \times L$ operator

- ❖ Image smoothed with 3×3 , 5×5 , 9×9 and 11×11 box filters



Original Image



3×3 box Filter

5×5 box Filter,

9×9 box Filter

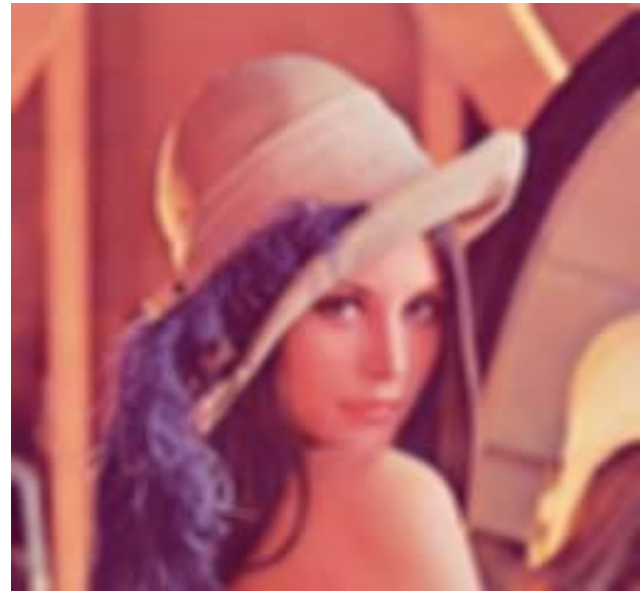
11×11 box filters

Linear Image smoothing techniques

Box filters. Arithmetic mean $L \times L$ operator



Original Lena image



Lena image filtered with
5x5 box filter

Linear Image smoothing techniques

- ❖ Computes a weighted average of pixels in the window
- ❖ Less blurring, less noise cleaning for the same size
- ❖ The family of binomial filters can be defined recursively

1-D Weighted filters

- ❖ The Simplest and most elementary smoothing mask is-

$$B = 1/2 [1 \ 1]$$

- ❖ It averages the gray value of two neighboring pixels. We can use this mask p times in a row on the same image. This corresponds to the filter mask

$$(1/2^P) [1 \ 1]^* [1 \ 1]^* [1 \ 1]^* \dots [1 \ 1]$$

- ❖ Written as an operator equation,

$$B^P = BBB \dots B$$

Linear Image smoothing techniques

$$\mathbf{b}^1 = \frac{1}{2} \begin{bmatrix} 1 & 1 \end{bmatrix}$$

$$\mathbf{b}^2 = \frac{1}{4} \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} = \mathbf{b}^1 * \mathbf{b}^1$$

$$\mathbf{b}^4 = \frac{1}{16} \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \end{bmatrix} = \mathbf{b}^1 * \mathbf{b}^1 * \mathbf{b}^1 * \mathbf{b}^1$$

$$\mathbf{b}^6 = \frac{1}{64} \begin{bmatrix} 1 & 6 & 15 & 20 & 15 & 6 & 1 \end{bmatrix} = \mathbf{b}^1 * \mathbf{b}^1 * \mathbf{b}^1 * \mathbf{b}^1 * \mathbf{b}^1 * \mathbf{b}^1$$

Linear Image smoothing technique

$$\mathbf{b}^4 = \frac{1}{4} \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} * \frac{1}{4} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$\mathbf{b}^8 = \frac{1}{16} \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \end{bmatrix} * \frac{1}{16} \begin{bmatrix} 1 \\ 4 \\ 6 \\ 4 \\ 1 \end{bmatrix} = \frac{1}{256} \begin{bmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 16 & 24 & 16 & 4 \\ 6 & 24 & 36 & 24 & 6 \\ 4 & 16 & 24 & 16 & 4 \\ 1 & 4 & 6 & 4 & 1 \end{bmatrix}$$

Weighted Filter

$$\frac{1}{16} \times$$

1	2	1
2	4	2
1	2	1

3*3 Weighted mask

Smoothing Spatial Filter

$$\frac{1}{16} \times$$

1	2	1
2	4	2
1	2	1

3*3 Weighted mask

$$g(x,y) = 1/16 \sum_{i=-1}^1 \sum_{j=-1}^1 W_{i,j} f(x+i, y+j)$$

Spatial Filter Generation Operation

$$g(x,y) = \frac{\sum_{i=-a}^a \sum_{j=-b}^b W_{ij} f(x+i,y+j)}{\sum_{i=-a}^a \sum_{j=-b}^b W_{ij}}$$

Mask Size = M X N

Where, M=2a+1, N=2b+1

Average Vs. Weighted Average



Original Image

$$\frac{1}{9} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$



$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$



Linear Image smoothing techniques

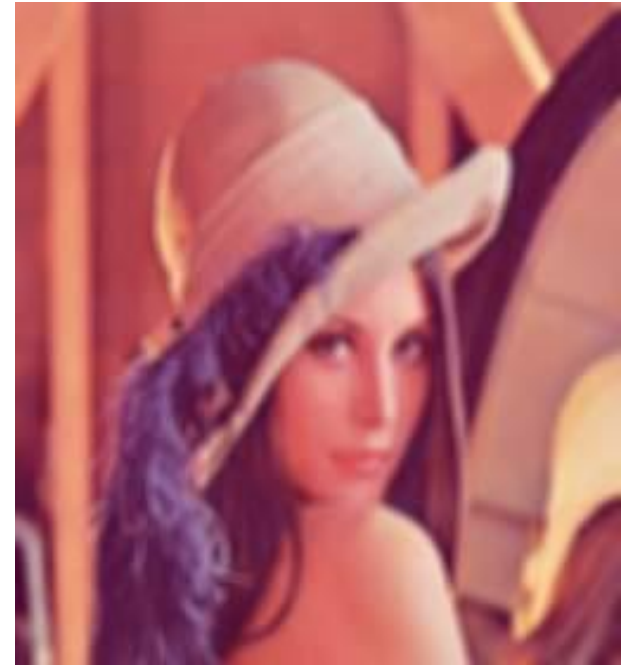
Weighted filters. Example



Original Lena image



**Lena image filtered
with Weighted 5x5 kernel**

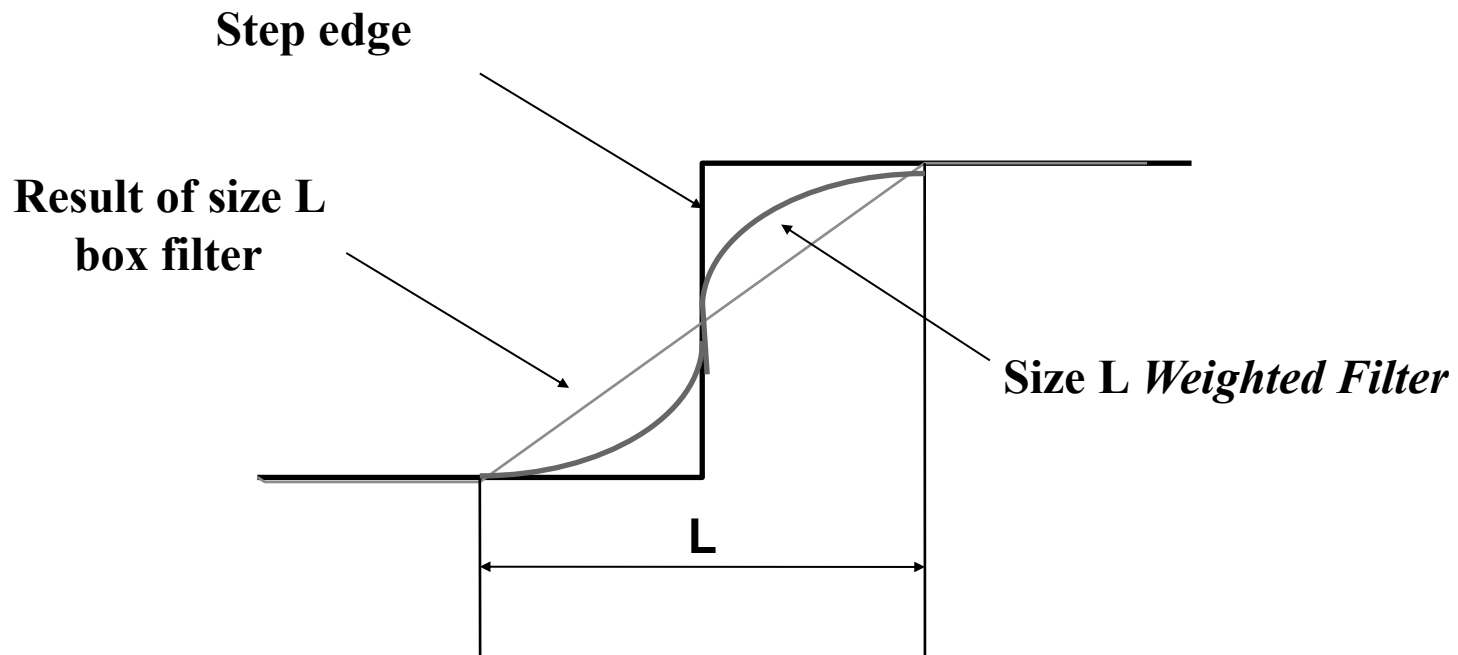


**Lena image filtered
with box filter 5x5**

Linear Image smoothing techniques

Weighted and box filters. Edge blurring comparison

- ❖ Linear filters have to compromise smoothing with edge blurring



Order-Statistics/Median Filters

- ❖ Order-statistics filters are nonlinear spatial filters whose response is based on ordering (ranking) the pixels contained in the image area encompassed by the filter, and then replacing the value of the center pixel with the value determined by the ranking result.
- ❖ The best-known example in this category is the *median filter*, which, as its name implies, replaces the value of a pixel by the median of the gray levels in the neighborhood of that pixel (the original value of the pixel is included in the computation of the median).
- ❖ Median filters are quite popular because, for certain types of random noise, they provide excellent noise-reduction capabilities, with considerably less blurring than linear smoothing filters of similar size.
- ❖ Median filters are particularly effective in the presence of *impulse noise*, also called *salt-and-pepper noise* because of its appearance as white and black dots superimposed on an image.

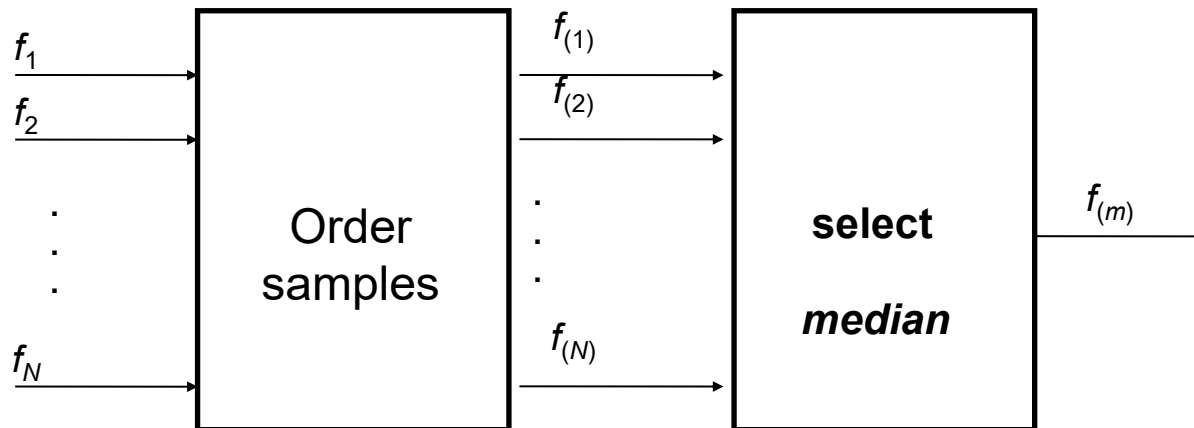
Median Filter

- ❖ The median of a set of values is such that half the values in the set are less than or equal to median value, and half are greater than or equal to median value.
- ❖ In order to perform median filtering at a point in an image, we first sort the values of the pixel in question and its neighbors, determine their median, and assign this value to that pixel.
- ❖ For example, in a 3×3 neighborhood the median is the 5th largest value, in a 5×5 neighborhood the 13th largest value, and so on.
- ❖ When several values in a neighborhood are the same, all equal values are grouped. For example, suppose that a 3×3 neighborhood has values(10, 20, 20, 20, 15, 20, 20, 25, 100).
- ❖ These values are sorted as (10, 15, 20, 20, 20, 20, 20, 25, 100), which results in a median of 20.

Nonlinear image smoothing

The median filter [Pratt 1991]

- Block diagram



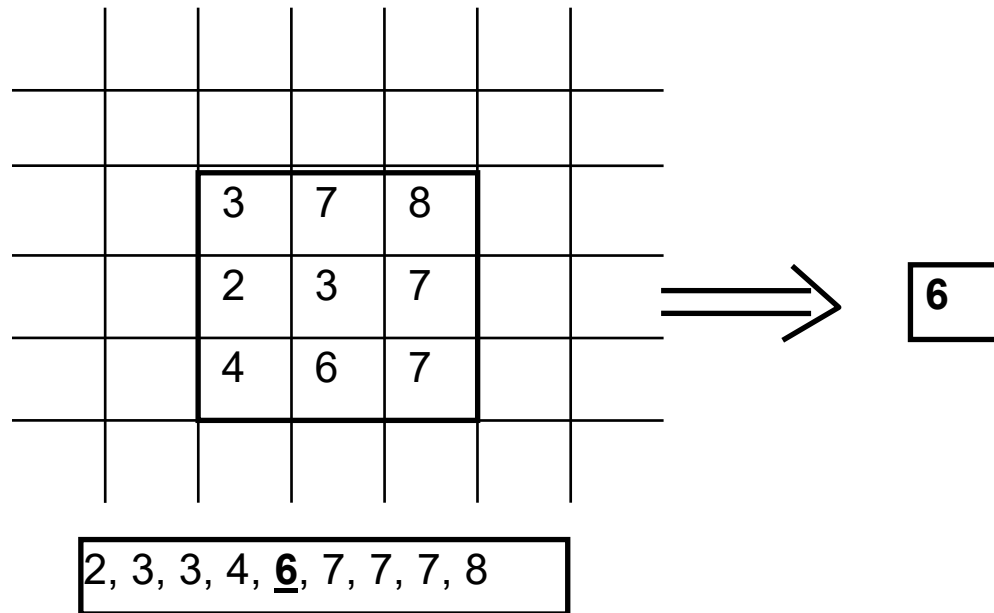
$$m = \frac{N+1}{2}$$

N is odd

Nonlinear image smoothing

The median filter

❖ Numerical example



Nonlinear image smoothing

The median filter

❖ It support Nonlinearity property.

➤ $\text{median}\{f1 + f2\} \neq \text{median}\{f1\} + \text{median}\{f2\}$.

However:

$$\text{median}\{c f\} = c \text{ median}\{f\},$$

$$\text{median}\{c + f\} = c + \text{median}\{f\}.$$

- The filter **selects** a sample from the window, does **not** average
- Edges are better preserved than with linear filters.
- Best suited for “salt and pepper” (Impulse noise) noise.

Nonlinear image smoothing

The median filter



Noisy image



5x5 median filtered



5x5 box filter

Nonlinear image smoothing

The median filter

- ❑ Implementing the median filter

- ❖ Sorting needs $O(N^2)$ comparisons

 - Bubble sort

 - Quick sort

 - Huang algorithm (based on histogram)

- ❑ **Caution: points, thin lines and corners are erased by the median filter**

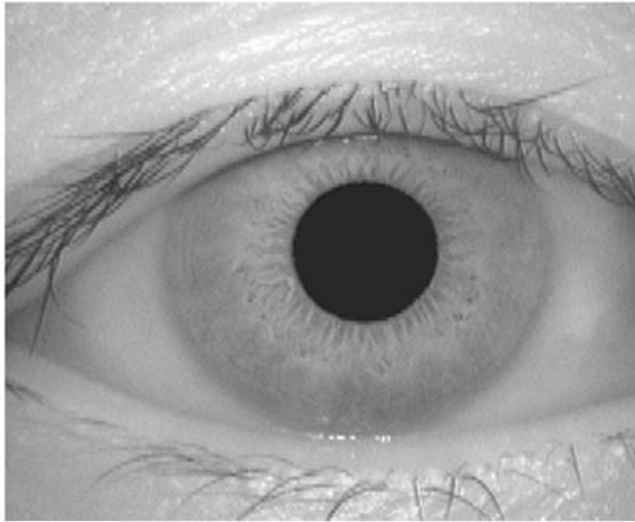
Nonlinear image smoothing

The median filter

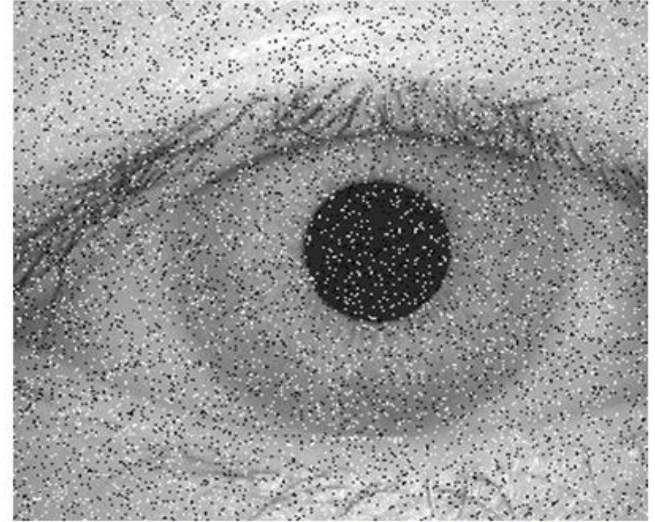
- Example of color median filtering
- 5x5 pixels window
- Image (a) original image
- Image (b) filtered image



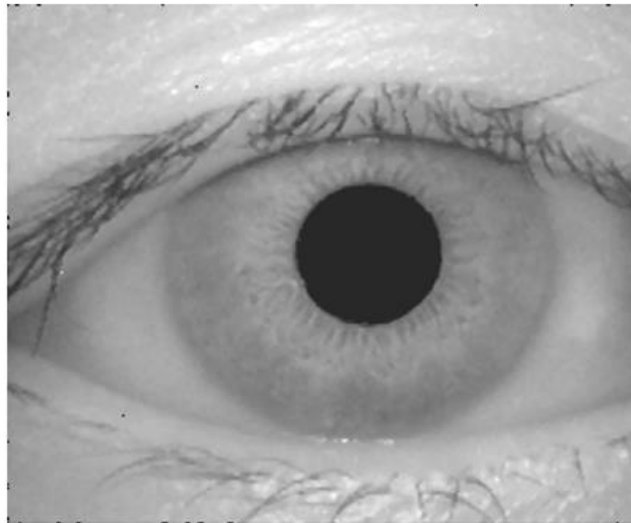
Median filter



Original Image



Corrupted Image



Median filtered Image

Drawback of Median Filter

- Removes Image Detail
- Signal Dependent Noise

Variations of Median Filter

- ❖ Weighted Median Filter
- ❖ Center Weighted Median Filter
- ❖ Adaptive Center Weighted Median Filter

Nonlinear image smoothing

The weighted median filter

- ❖ The basic idea is to give higher weight to some samples, according to their position with respect to the center of the window.
- ❖ Each sample is given a weight according to its spatial position in the window.
- ❖ Weights are defined by a weighting mask.
- ❖ Weighted samples are ordered as usually.
- ❖ The weighted median is the sample in the ordered array such that neither all smaller samples nor all higher samples can cumulate more than 50% of weights.
- ❖ If weights are integers, they specify how many times a sample is replicated in the ordered array.

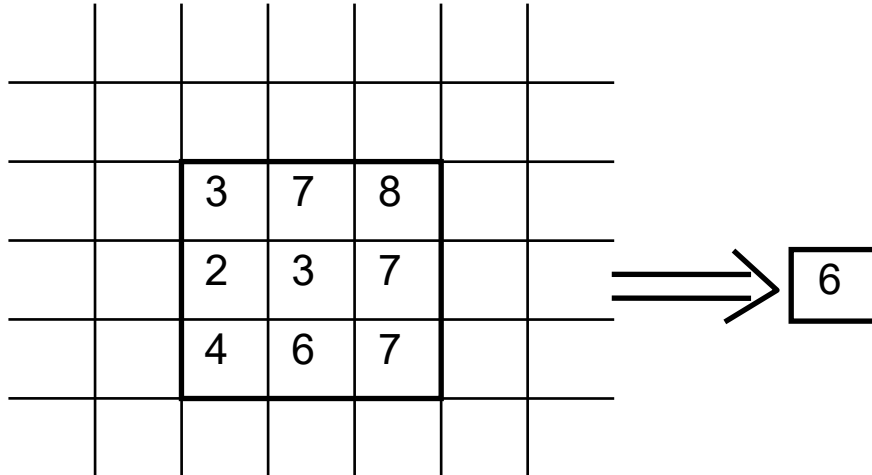
Nonlinear image smoothing

The weighted median filter

❖ Numerical example for the weighted median filter

1	2	1
2	3	2
1	2	1

Mask Filter Specify how many times a sample is replicated in the ordered array



2, 2, 3, 3, 3, 3, 4, 6, 6, 7, 7, 7, 7, 7, 8

Thanks